

# On the Non-Identifiability of Steering Vectors in Large Language Models



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THE CENTRAL CLAIM

Activation steering controls LLM behavior by adding a direction extracted from contrasting prompts — and is routinely read as evidence that a concept *lives along* that direction. That reading assumes the direction is uniquely recoverable from behavior.

It is not. An entire high-dimensional **family of directions** produces behavior indistinguishable from the original.

Row space

OBSERVABLE

$Jv \neq 0$

The only component steering can move. Everything you measure at the output lives here.

Null space

UNOBSERVABLE

$Jv_0 = 0$

Add any direction from here and not a single output changes. Infinitely many of them exist.

FOUR FINDINGS

- 1

Orthogonal twins are invisible

Random directions orthogonal to  $v$  leave behavior unchanged — mean Cohen's  $d$  of **0.119** (Llama) and **0.131** (Qwen).
- 2

Different methods, same behavior

Vectors from **cos -0.54** to **+0.32** — near-orthogonal — yet every pair stays at  $d < 0.27$ .
- 3

The class is high-dimensional

Across **50** orthogonal directions per trait, most traits sit near  $|d| \approx 0.08$ – $0.13$  — not a thin shell around  $v$ .
- 4

Stable, not an artifact

Holds across steering strength  $\alpha \in [0, 3]$  and survives topic- and genre-shift — a geometric property.

EXPERIMENTAL DESIGN

LLAMA-3.1 8B · GQA · L16/32

QWEN2.5 3B · GQA · L18/36

5 traits (formality, politeness, sentiment, truthfulness, agreeableness) · 50 contrastive prompt pairs each · effect measured as **Cohen's  $d$**  via pretrained classifiers · tests: orthogonal perturbation, method comparison, null-space spanning, multi-environment.

PERTURBATIONS STAY NEGLIGIBLE — Mean Cohen's  $d$ ,  $v$  vs  $v + v_\perp$  ·  $n = 10$  seeds

| TRAIT         | LLAMA-3.1 8B | QWEN2.5 3B |
|---------------|--------------|------------|
| Formality     | 0.059        | 0.151      |
| Politeness    | 0.159        | 0.120      |
| Sentiment     | 0.104        | 0.150      |
| Truthfulness  | 0.085        | 0.092      |
| Agreeableness | 0.186        | 0.156      |

— Vertical tick marks the  $|d| = 0.2$  negligible-effect threshold — all ten cells fall below it.

WHY DOES THIS HAPPEN?

The map from activations to logits,  $J = \partial o / \partial h$ , is rank-deficient in overparameterized models. Its null space has dimension  $d - \text{rank}(J) \geq 1$ , so infinitely many directions satisfy  $Jv_0 = 0$ .

Behavior pins down only the row-space shadow of  $v$  — never  $v$  itself.

$$J(v + v_0) = Jv + Jv_0$$
$$= Jv + 0$$

Identical logits, identical behavior — for any null-space direction  $v_0$ .

